

PROCESS FLOW DOCUMENT

Retail Demand Forecasting Using Walmart Store Sales Data

AI Engineer Technical Assignment – NEUZENAI IT SOLUTIONS PVT LTD

1. Introduction

This document explains the complete end-to-end pipeline developed for the Retail Demand Forecasting project using Walmart Store Sales data. The objective of this project is to forecast weekly retail sales across Walmart stores and departments using historical sales data, external economic indicators, promotional markdowns, and holiday information.

The solution combines:

- Time Series Forecasting
- Machine Learning Models
- Explainable AI (XAI)
- Business Intelligence
- Production-Oriented Pipeline Design

The process flow has been designed to ensure:

- Scalability
- Reproducibility
- Explainability
- Deployment readiness
- Business interpretability

2. End-to-End Pipeline Architecture



3. Data Ingestion

Objective

The purpose of data ingestion is to collect and integrate multiple data sources into a unified forecasting dataset.

3.1 Dataset Sources

The Walmart forecasting dataset contains multiple CSV files:

File Name	Purpose
train.csv	Historical weekly sales data
test.csv	Future forecasting dataset
stores.csv	Store metadata
features.csv	Economic indicators and markdown data
sampleSubmission.csv	Kaggle output reference

3.2 Data Loading

All datasets were loaded using the pandas library.

Libraries Used

```
import pandas as pd
```

Loading Workflow

```
train = pd.read_csv('train.csv')
```

```
features = pd.read_csv('features.csv')
```

```
stores = pd.read_csv('stores.csv')
```

3.3 Dataset Merging

The datasets were merged using:

- Store
- Date
- IsHoliday

This created a consolidated forecasting dataset containing:

- Historical sales
- Store metadata
- Economic indicators
- Promotional information
- Holiday effects

4. Data Validation

Objective

Ensure the quality and consistency of data before model development.

Validation Steps

Shape Validation

Checked dimensions of all datasets.

Null Value Analysis

Detected missing values across all features.

Datatype Validation

Validated numeric and categorical columns.

Duplicate Record Check

Ensured no duplicate rows existed.

Time Consistency Check

Verified chronological ordering of weekly data.

5. Data Cleaning & Preprocessing

Objective

Prepare clean and model-compatible data for forecasting.

5.1 Missing Value Handling

The Markdown columns contained significant missing values.

Strategy Used

Column Type	Missing Value Strategy
Markdown1-5	Filled with 0
CPI	Median Imputation
Unemployment	Median Imputation

Rationale

- Missing markdown values imply no promotional markdowns.
- Median imputation prevents distortion caused by outliers.

5.2 Date Processing

The Date column was converted into datetime format.

Extracted Features

Feature	Purpose
Year	Long-term trend analysis
Month	Seasonal patterns
Week	Weekly cyclic behavior
Quarter	Quarterly retail trends

Day	Temporal granularity
-----	----------------------

5.3 Categorical Encoding

Store type variables were encoded using one-hot encoding.

Example

Original	Encoded
Type A	Type_A
Type B	Type_B
Type C	Type_C

5.4 Numeric Conversion

All model input features were converted to numeric float format to ensure compatibility with:

- XGBoost
- SHAP
- LightGBM
- TensorFlow

6. Exploratory Data Analysis (EDA)

Objective

EDA was performed to understand patterns, trends, seasonality, and business behavior.

6.1 Sales Trend Analysis

Weekly sales trends were visualized across time.

Key Findings

- Strong seasonal spikes observed

- Holiday periods showed high demand
- Sales volatility differed across stores

6.2 Holiday Analysis

Holiday and non-holiday sales distributions were compared.

Findings

- Holiday weeks significantly increased sales
- Thanksgiving and Christmas showed the highest spikes

6.3 Store-Level Analysis

Store-wise sales distributions were analyzed.

Findings

- Larger stores generated higher revenue
- Store type influenced demand patterns

6.4 Correlation Analysis

A correlation heatmap was generated.

Key Correlated Features

- Lag features
- Rolling means
- Store size
- Markdown variables

7. Feature Engineering Pipeline

Objective

Generate meaningful predictive variables that improve forecasting performance.

7.1 Lag Features

Lag features help capture historical dependencies.

Features Created

Feature	Description
Lag_1	Previous week sales
Lag_4	Previous month sales
Lag_12	Previous quarter sales

Business Importance

Retail demand is strongly dependent on recent historical sales.

7.2 Rolling Window Features

Rolling statistics capture short-term trends and volatility.

Features Created

Feature	Description
Rolling_Mean_4	4-week average sales
Rolling_STD_4	4-week demand volatility

7.3 Holiday Features

A holiday proximity feature was created.

Feature

Feature	Description
Holiday_Next_Week	Indicates upcoming holiday demand

7.4 Interaction Features

Interactions between:

- Store size
- Promotions
- Holiday periods

were implicitly captured by tree-based models.

8. Train-Test Split Strategy

Objective

Prevent data leakage and simulate real-world forecasting.

Strategy Used

A time-series-aware split was implemented.

Dataset	Period
Training Set	Before 2012
Testing Set	2012 and after

Why Random Split Was Avoided

Random train-test splits are inappropriate for forecasting because:

- Future information may leak into training
- Temporal relationships break
- Unrealistic performance estimates occur

Time-series splitting preserves chronological integrity.

9. Model Development

Objective

Build multiple forecasting models and compare performance.

9.1 Random Forest Regressor

Advantages

- Handles non-linear relationships
- Robust against overfitting
- Easy baseline implementation

9.2 XGBoost Regressor

Advantages

- High forecasting accuracy
- Efficient boosting framework
- Excellent handling of structured retail data
- Strong feature importance capability

XGBoost achieved the best overall performance.

9.3 LightGBM Regressor

Advantages

- Faster training
- Lower memory consumption
- Good scalability for large datasets

9.4 Prophet Forecasting Model

Facebook Prophet was implemented for:

- Trend decomposition
- Seasonal analysis
- Holiday-aware forecasting

10. Hyperparameter Tuning

Objective

Optimize model performance and reduce overfitting.

Technique Used

RandomizedSearchCV was applied to XGBoost.

Tuned Parameters

Parameter	Purpose
n_estimators	Number of trees
max_depth	Tree complexity
learning_rate	Learning speed

Benefits

- Improved generalization
- Reduced overfitting
- Better forecasting stability

11. Model Evaluation

Objective

Evaluate forecasting accuracy and robustness.

Evaluation Metrics

Metric	Purpose
--------	---------

MAE	Average prediction error
RMSE	Penalizes large errors
WMAE	Holiday-weighted accuracy
MAPE	Percentage error
R² Score	Goodness of fit

Why WMAE Is Important

Retail businesses prioritize holiday accuracy because forecasting errors during holidays can:

- Cause stock shortages
- Increase inventory costs
- Reduce customer satisfaction

Holiday weeks were weighted 5x as specified in the Kaggle competition.

12. Residual Analysis

Objective

Analyze prediction errors for stability and robustness.

Analyses Performed

- Residual distribution plots
- Residual scatter plots
- Error randomness analysis
- Distribution normality checks

Findings

- Residuals were approximately centered around zero
- No strong systematic patterns observed
- Model predictions were stable across time

13. Time-Series Cross Validation

Objective

Evaluate model stability across multiple temporal folds.

Technique Used

TimeSeriesSplit cross-validation.

Benefits

- Prevents temporal leakage
- Simulates real-world forecasting
- Provides robust performance estimation

14. Robustness Testing

Objective

Evaluate model performance under different retail conditions.

Tests Conducted

Holiday vs Non-Holiday Weeks

Measured forecasting accuracy during demand spikes.

Store-Type Analysis

Compared performance across store categories.

Markdown Conditions

Evaluated forecasting under promotional campaigns.

15. Explainable AI (XAI)

Objective

Improve transparency and business trust in AI-driven predictions.

15.1 SHAP Explainability

SHAP was used for:

- Global feature importance

- Feature contribution analysis
- Dependence visualization

Important Features Identified

- Lag_1
- Rolling_Mean_4
- Store Size
- Markdown variables
- Holiday indicators

15.2 LIME Explainability

LIME was used for local prediction explanations.

Benefits

- Human-readable explanations
- Local decision transparency
- Stakeholder trust improvement

16. Business Insights

Key Findings

Holiday Impact

Holiday periods strongly increased demand.

Seasonal Demand

Retail demand followed clear seasonal trends.

Promotional Effects

Markdown campaigns directly influenced purchasing behavior.

Store Variability

Larger stores experienced higher demand volatility.

Economic Influence

Fuel prices and unemployment impacted customer spending.

17. Business Recommendations

Inventory Planning

Increase inventory before holidays and promotional periods.

Workforce Optimization

Use forecasts for staffing and workforce planning.

Markdown Optimization

Plan promotions using forecasting insights.

Store-Specific Forecasting

Develop localized forecasting systems.

Strategic Planning

Use explainable AI insights for transparent decision-making.

18. Model Fairness Discussion

Objective

Evaluate whether forecasting performance remains equitable across store categories.

Observations

- Model performance remained relatively stable across store types.
- Holiday-related forecasting errors were slightly higher.
- Smaller stores with limited historical data showed slightly lower accuracy.

Potential Biases

- Data imbalance across stores
- Extreme markdown events
- Regional sales variations

Future Fairness Improvements

- Store-specific ensemble models
- Region-aware forecasting

- Fairness-aware optimization strategies

19. Production-Ready Pipeline

Objective

Design a scalable and deployable forecasting solution.

Production Features

- Modular architecture
- Reproducible workflow
- Organized project structure
- Explainability integration
- Model persistence
- Scalable forecasting pipeline

20. Future Enhancements

Potential improvements include:

- Real-time forecasting APIs
- Streamlit dashboard deployment
- Cloud deployment
- MLflow experiment tracking
- Ensemble forecasting
- Deep learning optimization
- Automated retraining pipelines

21. Conclusion

This project successfully developed a comprehensive retail demand forecasting pipeline using Walmart sales data.

Multiple forecasting approaches were implemented and compared using robust evaluation techniques. XGBoost achieved the best overall forecasting performance due to its strong predictive capability, scalability, and robustness.

Explainable AI methods such as SHAP and LIME improved model transparency and business interpretability, making the solution suitable for real-world retail analytics and AI-driven decision-making.

The project demonstrates practical expertise in:

- Machine Learning
- Time Series Forecasting
- Explainable AI
- Retail Analytics
- Production-Oriented AI Engineering